Internship Report – Day 11

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 11 Overview:

Today, I trained an Automatic Number Plate Recognition (ANPR) model. The dataset included vehicle images with annotated license plates. The model learned to detect and extract number plates using object detection techniques. It will now be tested for accuracy and optimized for real-time use.

Note:- I am still training the model for more accuracy its taking much time. So I have started learning the blender to create the 3D models .

Tools Used Today

GEMINI, SONNET, Chatgpt, vs coder.

Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VECHILES NUMBERPLATES.

Approaches:-

The ANPR system was approached using deep learning-based object detection models like YOLO or SSD. Preprocessing included resizing images and annotating number plate regions. The model was trained to detect and localize plates in diverse lighting and angle conditions. Post-processing involves OCR to extract text from the detected plate regions.

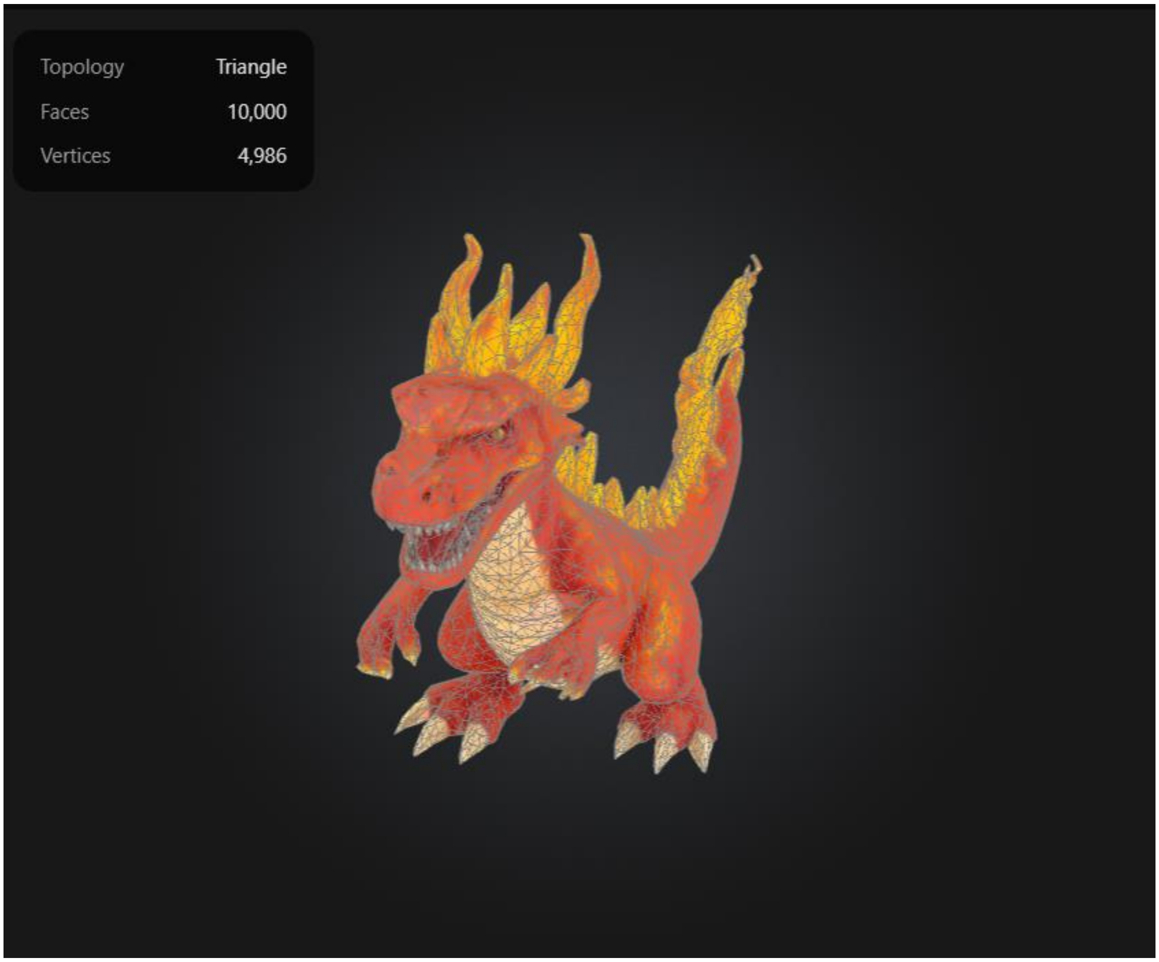
Attachments:-

```
|          | 12/19 [01:26<00:49,  7.11s/it]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):  68%|          | 13/19                 Class     Images  Instances      Box(P
AP50  mAP50-95):  74%|          | 14/19 [01:40<00:35,                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):  79%|          | 15/19 [01
       Class     Images  Instances      Box(P          R      mAP50  mAP50-95):  84%|          | 16/19 [01:54<                 Class     Images  Instances      Box(P
50  mAP50-95):  89%|          | 17/19 [02:16<                 Class     Images  Instances          R      mAP50  mAP50-95):  95%|          | 18/19 [02:40<
ges  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 19/19 [02:42<                 Class     Images  Instances      Box(P          R      mAP50  mA
          | 19/19 [02:42<00:00,  8.56s/it]
                   all        579        594       0.54      0.533      0.533      0.278
               licence        579         50      0.359       0.12      0.147     0.0464
                 plate        579        544      0.721      0.947      0.918       0.51

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      3/100         0G      1.627      1.022      1.535
                 Class     Images  Instances      Box(P
                 Class     Images  Instances      Box(P
                   all        579        594      0.722       0.64      0.657       0.34
               licence        579         50      0.573       0.32      0.349      0.127
                 plate        579        544      0.872       0.96      0.965      0.553
```

I am training the again to get the more accuracy with the 100 epochs.

Other than this training this model I have learned about the blender to create the 3D model .



This is the 3d model I have created by using the AI while the model is training.